6 Everyday use and help

FULL

10. The plan: a careful order, with checkpoints

Quick version

No time to read it all? This is the short version. The full guide is below.

- A careful order for the guides, with a checkpoint after each step. A "session" is about 2 to 3 hours.
- Build the foundation first and prove each layer works. Never add a new thing on top of something that is not working.
- Before you begin (15 minutes): read guide 00, start a notes file `lab-notes.txt`, and have a password manager ready.
- The order: Ubuntu, then Docker and your garden, Uptime Kuma and backups, Gemini, Home Assistant, n8n email, sensors and watering, your readings on the website, and last the internet.
- The internet steps (sessions 14 to 16): practise with a test name first (guide 04, Part C), then do it for real with guide 23, Parts 9 to 11.
- Watering: the valve must work dry on the table before it ever touches water. Never skip the dry test.
- Every week (15 minutes): `bash lab/09-health-check.sh`, Ubuntu updates, and a copy of `backups/` on a USB stick.

This is the careful, step-by-step plan. It puts the guides in the best order, says when to stop and check, and keeps risky steps for later. Go at your own speed. A "session" below is about 2 to 3 hours. You can split one over two days.

The idea: build the foundation first, prove each layer works, then add the next layer. Never add a new thing on top of something that is not working.

Before you begin (15 minutes)

- [ ] Read 00 Read me first (the 3 rules).
- [ ] Open a notes file called `lab-notes.txt`. Each session, write the date, what you did, and what you saw. This helps you and Gemini.
- [ ] Have a password manager ready. Every new password goes in it first.
- [ ] Charger plugged in. Laptop on a table, not a bed or a couch.

Session 1. A clean base

Guides: 01 Ubuntu, then `bash lab/01-prepare-ubuntu.sh`. **Checkpoint:** `bash lab/09-health-check.sh` shows no red FAIL for the laptop section. The laptop restarts and comes back. **Why careful:** a good base saves hours later.

Session 2. Docker and your garden

Guides: 02 Docker (or `bash lab/02-install-docker.sh`), then 03 Run your garden (or `bash lab/03-start-garden.sh`). **Checkpoint:** - `curl http://127.0.0.1:8030/healthz` says ok. - You logged in and added one plant. - You took a backup (menu `6`) and the file exists. **Stop here for a day.** Look at the site. Read `rust/junsgarden/src/main.rs`.

Session 3. Watch it and protect it

Guides: Project 1 Uptime Kuma, Project 2 Backups. **Checkpoint:** you stopped the site on purpose and Uptime Kuma noticed. You restored a deleted test plant from a backup. **Why careful:** you should know your safety net works before you rely on it.

Session 4. Your helper (Gemini)

Guide: 08 Gemini setup. **Checkpoint:** your Gem answers in numbered steps, and you ran `bash lab/10-collect-for-gemini.sh` and read the file it made.

Sessions 5 to 7. Home automation

Guides: Project 3 Home Assistant, then Project 4 Govee, then Project 5 Weather. **Checkpoint after each:** the thing works twice in a row, after a restart too. **Stop and enjoy it.** A light that comes on at sunset is a real win. Tick it in MY-WINS.md.

Sessions 8 to 9. Email and automation with n8n

Guides: n8n 1, n8n 2, n8n 3. **Checkpoint:** you received emails from your own lab at `you@gmail.com`.

Sessions 10 to 12. Sensors and watering (hardware)

Guides: Project 6 Soil sensor, then Project 7 Watering. Buy parts only when you reach a project (shopping list). **Checkpoints:** the sensor changes when you move it from air to water. The valve works **dry on the table** before it ever touches water. The watering stops by itself at the time limit. Ask a friend who knows electrics to check the wiring before the first power-on. **Never skip the dry test.**

Session 13. Your readings on your website

Guide: Project 8. **Checkpoint:** a new row in the database every 15 minutes.

Sessions 14 to 15. The internet: practise first

Guides: 23 The full setup, Part 9 and 04 Internet with Caddy, Apache and No-IP, Parts A to C: your own No-IP account and web name, the CGNAT check, then a **test name** (menu `M`, `1`, choice `2`, for example `garden.test`). The scripts `lab/04`, `lab/05` and `lab/06` do the same jobs one at a time, if you want to learn each piece. **Checkpoint:** on the garden computer, `https://garden.test` opens your site (Firefox warns about the practice certificate: that is expected), and `/beds` opens after you sign in. Your web name is in your own No-IP account, with its DDNS key in your password manager. **Why careful:** the next session is the only step that opens your home network to the internet. Practise until it feels familiar, and do the real thing on a calm day with time to spare.

Session 16. Your web name for real (junsgarden.zapto.org)

Guides: 23 The full setup, Parts 10 and 11, 04, Part D, and 04b. Menu `M`, `1`, choice `3`, then forward only ports 80 and 443 in the router (guide 14). **Checkpoint:** from a phone on mobile data, the real name opens your site with a padlock, with the real plants, and you can log in. Run `bash lab/09-health-check.sh`: no FAIL. You know the way back: menu `M`, then `14` takes the garden off the internet again (guide 04).

After that. Keep it healthy (15 minutes a week)

- [ ] `bash lab/09-health-check.sh` and fix any FAIL.
- [ ] `sudo apt update && sudo apt upgrade -y`, then `sudo reboot` if asked.
- [ ] Copy `backups/` to a USB stick.
- [ ] Look at Uptime Kuma. Is everything green?

Later and optional

- 07 An SSO sign-in gate for your tools.

- Project 9 More ideas.

Pace and rest

- Take a break every hour. Stand up, drink water.
- If you feel stuck or tired, stop and come back. Nothing here is a race.
- One new thing at a time. Finishing small things is better than starting many.
- Write down what you want to learn next in `lab-notes.txt`. Your list matters.

Back to 00 Read me first.